Billing & Warehouse
Operating Workflow

One application, two modes — from a single counter to a multi-warehouse operation

This document describes how the system is actually operated, end to end. It covers the Basic workflow — the one a single shop runs every day — and the Advanced workflow that a multi-location or third-party warehouse business runs on the same database, with the same records, when it grows into it.

It is deliberately a description of the working system rather than a specification of an intended one. Every step below is exercised by the automated workflow suite, so a step that stopped being true would show up as a failing test rather than as a document quietly drifting away from the software.

> **THE GOVERNING PRINCIPLE**
> There is one application, one database and one set of business rules. Basic and Advanced are not separate products or separate installations — they are the same system with different modules switched on. A shop that grows does not migrate; it turns things on, and its existing stock, customers, ledgers and history carry straight over.

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Regenerate with: npm run docs:workflow

The two modes

Business mode decides which modules are available. Basic modules are always on. Advanced modules appear only in Advanced mode, and each can still be switched off individually — so a warehouse business that has no interest in double-entry bookkeeping simply does not have it.

Capability	Basic	Advanced
Counter billing, invoices, returns	Yes	Yes
Products, customers, suppliers, purchases	Yes	Yes
Customer & supplier ledger (udhar / khata)	Yes	Yes
Expenses, cash register, cash flow	Yes	Yes
Batch & expiry tracking	Optional	Yes
Stock audit and reconciliation	Yes	Yes
Warehouses, zones, aisles, racks, bins	—	Yes
Purchase orders and goods receipt (GRN)	—	Yes
Put-away, picking, packing, dispatch	—	Yes
Stock transfers between locations	—	Yes
Serial number tracking	—	Yes
Approval workflow	—	Yes
Double-entry accounting and financials	—	Yes
Third-party (3PL) stock and storage billing	—	Optional

SWITCHING MODES IS SAFE AND REVERSIBLE

Moving to Advanced does not rewrite anything. Stock, ledgers and history are untouched; new capabilities simply become reachable. Switching back closes those doors again and leaves the data intact. This is verified in both directions.

Basic workflow — the shop day

This is the whole operating cycle for a single-counter business: set up what you sell, buy it in, sell it, take the money, record what it cost you, and check at the end that the numbers agree with themselves.

Set up » Buy in » Sell » Take payment » Record costs » Close the day » Check

Setting up what you sell

1. **Create categories and products**
   A product carries its purchase price, selling price, MRP and GST rate. It can also carry a secondary unit — for example 1 BOX = 10 PKT — which lets you buy in boxes and sell in packets without doing arithmetic in your head.

2. **Add suppliers and customers**
   A customer added at the counter mid-bill is the common case, so the invoice screen can create one inline without leaving the sale.

> **MRP IS ENFORCED AT ENTRY**
> Selling above a printed MRP is refused when the product is saved, not discovered later on a bill. A secondary unit without a real conversion factor is refused for the same reason — it looks like a working setup while silently billing the wrong quantity.

Goods coming in

3. **Record a purchase**
   Entered in whichever unit the supplier invoiced. Buying 5 BOX of a product whose conversion is 10 adds 50 packets to stock, and the purchase, the stock movement and the supplier ledger entry are written together.

Selling

4. **Raise an invoice, or bill at the counter**
   Cash, card, UPI, bank transfer or credit. Stock falls as the bill is saved, and the movement is written in the same transaction — a sale can never reduce stock without leaving a trace, or leave a trace without reducing stock.

5. **Selling more than you have is refused**
   Unless the company has deliberately allowed negative stock. Two counters selling the last unit at the same moment resolve to one sale and one clear refusal, not two sales and a mystery.

Money

6. **Record payments against the bill**
   Part payments are normal. What a customer owes is always derived from their ledger rather than stored on their record — a cached balance is the thing that goes stale and gets argued over.

7. **Udhar / khata**
   The running account for a party, showing every bill and payment in order, with the closing balance at the bottom.

Costs and the day's cash

8. **Record expenses**
   Rent, electricity, wages, repairs. These are Basic-mode features on purpose: the smallest shop tracks them, and only double-entry bookkeeping is genuinely advanced.

9. **Open the till in the morning, close it at night**
   Opening float in, closing count out, with the difference visible rather than absorbed.

Checking

10. **Reconcile stock**
    Three things should agree about every product: the balance, the sum of its movements, and the sum of its lots. The audit reports any that do not, valued at cost, so the largest problem is dealt with first.

11. **Reports and alerts**
    Sales, stock valuation and GST reports. The notification bell raises low stock, expiring lots, overdue receivables and any drift the audit found.

Advanced workflow — the warehouse

Everything in the Basic workflow still applies. What follows is what the same company gains when it operates a real warehouse: a physical layout, a controlled buying process, and a fulfilment cycle where goods are tracked from the receiving bay to the customer's door.

The building

A location is described as a tree, and every rung is optional — a small godown can stop at "Zone A" and never define a bin.

Warehouse » Zone » Aisle » Rack » Shelf » Bin

1. **Define the layout, then generate a walking route**
   Each bin gets a pick sequence: its position in the order a picker walks past it. The generated route is a serpentine — down the first aisle, back up the second — which is what a picker does naturally. Sequences are numbered with gaps so a new bin can be inserted later without renumbering the building.

> **THE TREE AND THE ROUTE ANSWER DIFFERENT QUESTIONS**
> The tree says what contains what. It cannot say that the last rack of aisle 1 is next to the last rack of aisle 2, because they sit in different branches. Picking by tree order therefore walks the length of the building once per aisle. The pick sequence is the authority on routing, and it is set by whoever knows the building.

Buying: purchase order to shelf

PO raised » Approved » GRN received » Posted to stock » Put away

2. **Raise and approve a purchase order**
   Orders below the approval threshold clear on submission; larger ones wait for a decision. Approval rules are configurable per document type and value.
3. **Receive goods against it (GRN)**
   Ordered, received and accepted quantities are recorded separately, so a short or damaged delivery is visible rather than averaged away.
4. **Post the receipt**
   A saved GRN has not moved stock. Posting it is the deliberate act that does, which means a receipt can be checked and corrected before it affects anything.
5. **Put away**
   Posted stock lands in the receiving bay and appears on the put-away queue until it is on a shelf. Put-away rules suggest where each product should go — cold goods to cold storage, fast movers near dispatch — but only suggest: the storeman can always choose otherwise, because a rule pointing at a full bin must not stop the work.

Selling: order to doorstep

Order » Allocate » Route » Pick » Pack » Dispatch » Delivered

6. **Allocate**
   Sets stock aside for the order. Nothing physical happens — what changes is that the goods are now spoken for, which is what stops two orders promising the same last box to two customers. A partial allocation is normal and is reported as such.

7. **Produce the pick list**
   Which bin to walk to for each line, oldest lot first, then ordered into a single walk across the whole order — not one walk per product.
8. **Release the route as tasks**
   Each stop becomes a warehouse task with the bin, product, quantity and walk position on it. From here the work is assignable, measurable and can be handed over at shift change.
9. **Pick, then pack**
   Picking takes goods off the shelf onto the packing bench. Packing puts them into cartons, each with its own package number and weight.
10. **Dispatch, then track delivery**
    Courier and tracking number are recorded, and the order can be followed through to delivered.

> **STOCK LEAVES THE LOCATION EXACTLY ONCE, AT DISPATCH**
> Allocation, picking and packing all happen inside the building, so the location total must not move. Only dispatch takes goods out. This is why the shelf figure does not change when an order is picked — and why a cancelled pick can put everything back without anything needing to be unwound.

> **EXPIRY DECIDES WHAT IS PICKED; THE ROUTE DECIDES ONLY THE ORDER**
> Oldest lot first is an inventory rule and is not negotiable. The walking route is applied afterwards, to picks that have already been chosen. Folding the two together would let a shorter walk quietly outrank the expiry rule, and the first anybody would know is a write-off.

Moving and correcting stock

11. **Transfer between locations**
    Raised, approved, dispatched from one location and received at the other. Both legs are written to the ledger, and goods in transit are visible as such rather than missing.
12. **Adjustments and counts**
    Damage, expiry and counting differences are recorded as adjustments with a reason. Cycle counts can be run per location without stopping the warehouse.

Exceptions and labour

13. **Raise an exception when something is wrong**
    Short picks, over-receipts, damaged stock, wrong bin, wrong product, stock mismatch, expired batch, missing scan. Raising one is quick and never blocks the picker — forcing them to stop and reconcile means, in practice, that they stop reporting.
14. **Resolve it with an account of what was done**
    Closing an exception requires a resolution note. "Resolved" with no explanation is indistinguishable from ignored, except that it also hides the problem.
15. **Measure the work**
    Because every job is a task with a start and finish time, productivity per person and per task type falls out without anybody filling in a timesheet.

Third-party (3PL) storage

Optional, and off unless switched on. It lets the warehouse hold goods belonging to other companies alongside its own.

16. **Register the client and receive their goods**
    Their stock is held against them, at the same locations and in the same bins as yours, with a rate card for storage and handling.
17. **Capture storage daily, bill monthly**
    A snapshot of what was held is written every day by a background job. The monthly bill is the sum of those days.

> **A CLIENT'S GOODS ARE NEVER YOURS**
> Client stock is not sellable, not in your valuation, not in your catalogue total and not pickable on your orders. The separation is structural — a separate balance — rather than a filter somebody has to remember to apply.

> **STORAGE CHARGES ARE NEVER RECALCULATED FROM CURRENT STOCK**
> Goods that arrived on the 3rd and left on the 11th are invisible in a month-end balance, yet eight days are owed on them. Each day is written down while it is still true and never recomputed, so the same period billed twice gives the same figure.

The books

18. **Double-entry accounting**
    Sales, purchases, payments and expenses post to a chart of accounts automatically. Trial balance, profit and loss, and balance sheet are derived from the journal — never from stored totals — so they cannot disagree with it.

The rules that hold it together

These are the invariants the system is built around. They are listed because each one is easy to break by writing perfectly reasonable code, and because breaking any of them produces a problem that surfaces weeks later as unexplainable drift.

Rule	Why it matters
Stock only moves through one engine	Quantity and ledger entry are written together, in one transaction. A movement that changed stock without leaving a trace is the one bug an inventory system cannot recover from.
Bin quantities never exceed location stock	Bins are a sub-allocation of the location, not a second copy. Anything that takes goods out of a location releases them from their shelves too — enforced centrally, because the next outbound path somebody writes would forget.
The difference is the receiving bay	Stock at a location but in no bin is goods that have arrived and not yet been put away. Real warehouses have one, so the model does too.
Ownership is a dimension of stock	Every balance belongs to exactly one owner. A shop has one — itself — and never notices. Quantities never mingle across owners.
Catalogue stock and valuation are house-only	Counting a client's goods would silence reorder alerts on things you have none of, and put someone else's inventory on your balance sheet.
Statements are derived, never stored	Balances, trial balance and financials are computed from their underlying entries so they cannot drift away from them.
A repeated request happens once	Scanner operations carry an idempotency key. The unique index arbitrates, so two copies of a request cannot both do the work.
A task completes once	Guarded in the database, not in application logic. Two devices racing to finish one job resolve to one completion and one stock movement.
Deletion is soft, and audited	Records are marked rather than removed, with who and when, so history stays reconstructible.

Operating notes

Daily

A background job captures the previous day's storage position and sweeps expired idempotency keys. It runs on its own and catches up any days a switched-off server missed. It is safe to re-run: a duplicate day collides rather than billing twice.

Before invoicing a storage period

Check for missing snapshot days. A bill built over a period with gaps is quietly short, and the only moment anybody would notice is when the client queries it. The bill reports its own completeness for this reason.

Before printing pick routes

Check the route health for unsequenced bins. A bin with no pick sequence still works — it is visited at the end rather than failing the pick — but it means part of the layout has not been thought about.

First run against an existing database

Take a backup. The first boot after an upgrade applies schema changes, attributes existing stock to the house owner, and renames any duplicate bin codes within a warehouse so the uniqueness rule can be enforced. Every rename is logged so the shelf label can be corrected.

Roles

Role	Typically does
Admin	Everything, including settings, modes and modules.
Accountant	Books, approvals, financial reports, storage billing.
Warehouse Manager	Layout, routes, task assignment, exceptions, counts.
Inventory Staff	Put-away, picking, packing, raising exceptions.
Purchase Manager	Purchase orders, goods receipt, suppliers.
Sales	Counter billing, invoices, customers, quotations.
Branch Manager	One location's trading, cash and transfers.
Auditor	Read-only across stock audit and reports.

Rights are also granted per location, at three levels: View, Operate and Manage. An area manager can read three branches while billing at only one.